

# EchoVoice – Structured English

## LEVEL 1 PROCESSES

### Process 1.0 – Manage Exercise

```
BEGIN
    READ Active Goals FROM D2 Goal Config Store
    GET Exercise Selection FROM Caregiver

    IF Exercise Selection IS VALID AND CONSISTENT WITH Active
Goals THEN
        DETERMINE Target Word FOR the selected exercise
        SEND Target Word TO Process 2.0 (Capture Speech)
    ELSE
        REJECT selection
        PROMPT Caregiver TO choose a valid exercise
    ENDIF
END
```

### Process 2.0 – Capture Speech

```
BEGIN
    RECEIVE Target Word FROM Process 1.0
    DISPLAY/PLAY Target Word TO Learner (prompt)
    WAIT FOR Speech Input FROM Learner

    IF Speech Input IS RECEIVED WITHIN timeout THEN
        RECORD Speech Input AS Recorded Speech
        SEND Recorded Speech TO Process 3.0 (Speech Recognition
& Assessment)
    ELSE
        FLAG no-response event
        RETURN control TO Process 1.0
    ENDIF
END
```

### Process 3.0 – Speech Recognition & Assessment

```
BEGIN
    RECEIVE Recorded Speech FROM Process 2.0
```

```
PROCESS Recorded Speech (see Level 2 decomposition: 3.1-3.4)
PRODUCE Assessment Result

WRITE Assessment Record TO D1 Progress Database
SEND Assessment Result TO Process 4.0 (Generate Feedback)
END
```

#### **Process 4.0 – Generate Feedback**

```
BEGIN
    RECEIVE Assessment Result FROM Process 3.0

    IF Assessment Result INDICATES correct/acceptable
pronunciation THEN
        GENERATE positive Visual & Auditory Prompts
    ELSE IF Assessment Result INDICATES minor errors THEN
        GENERATE corrective Visual & Auditory Prompts
highlighting the error
    ELSE
        GENERATE encouraging retry Visual & Auditory Prompts
    ENDIF

    SEND Visual & Auditory Prompts TO Learner
END
```

#### **Process 5.0 – Monitor Progress**

```
BEGIN
    READ Stored Progress FROM D1 Progress Database

    COMPUTE Session Progress Summary FOR current session
    SEND Session Progress Summary TO Caregiver

    COMPUTE Progress Report (longitudinal trends across
sessions)
    SEND Progress Report TO SLP
END
```

#### **Process 6.0 – Configure Goals & Review Assessment**

```
BEGIN
```

```
RECEIVE Goal Configuration AND Assessment Oversight input
FROM SLP
```

```
IF input TYPE = Goal Configuration THEN
    VALIDATE goal parameters
    WRITE Goal Configuration TO D2 Goal Config Store
ELSE IF input TYPE = Assessment Oversight THEN
    REVIEW assessment data
    (no store update; used for oversight/reporting only)
ENDIF
END
```

## **LEVEL 2 – DECOMPOSITION OF PROCESS 3.0**

### **Process 3.1 – Extract Acoustic Features**

```
BEGIN
    RECEIVE Raw Audio File (.wav) FROM Process 2.0 (Capture
Speech)
    APPLY signal processing (e.g., framing, windowing,
transformation)
    PRODUCE Spectrogram Tensors
    SEND Spectrogram Tensors TO Process 3.2
END
```

### **Process 3.2 – Run Model Inference**

```
BEGIN
    RECEIVE Spectrogram Tensors FROM Process 3.1
    RUN on-device speech recognition model ON Spectrogram
Tensors
    PRODUCE Predicted Phoneme Sequence
    SEND Predicted Phoneme Sequence TO Process 3.3
END
```

### **Process 3.3 – Align Phonemes**

```
BEGIN
    RECEIVE Predicted Phoneme Sequence FROM Process 3.2
    READ Target Phoneme Template FROM D3 Phoneme Target
Dictionary
```

```

    ALIGN Predicted Phoneme Sequence WITH Target Phoneme
Template
    FOR EACH phoneme position DO
        IF predicted phoneme ≠ target phoneme THEN
            RECORD mismatch
        ENDIF
    ENDFOR

    PRODUCE Phoneme Error Matrix
    SEND Phoneme Error Matrix TO Process 3.4
END

```

### **Process 3.4 – Compute Pronunciation Score**

```

BEGIN
    RECEIVE Phoneme Error Matrix FROM Process 3.3
    CALCULATE pronunciation score FROM error matrix (e.g.,
weighted accuracy per phoneme)
    PRODUCE Assessment Result

    SEND Assessment Result TO Process 4.0 (Generate Feedback,
Level 1)
    WRITE Assessment Record TO D1 Progress Database
END

```

### **Data Stores Referenced**

**D1 – Progress Database:** stores Assessment Records; source of Stored Progress for monitoring.

**D2 – Goal Config Store:** stores Goal Configuration; source of Active Goals for exercise selection.

**D3 – Phoneme Target Dictionary:** source of Target Phoneme Templates for alignment.

### **External Entities**

**Learner:** provides Speech Input; receives Visual & Auditory Prompts.

**Caregiver:** provides Exercise Selection; receives Session Progress Summary.

**SLP (Speech-Language Pathologist):** provides Goal Configuration & Assessment Oversight; receives Longitudinal Progress Reports.